



Activity File: Restoring Data with Incremental Backups

In this activity, you will continue as a junior administrator at **Rezifp Pharma Inc**, which has just adopted a new backup policy.

You will be responsible for validating the new backup process by performing the following two tests:

- Use a backup to restore lost patient files.
- Create two new files and perform an incremental backup.

You will use the following process to test the new restoration procedure:

- List and verify the existing files in the `testenvir` directory.
- Create a **level 0** backup on Sunday containing the entire `testenvir` directory.
- Simulate a cyberattack by deleting the `patient` directory from the `testenvir` parent folder.
- Verify the `patient` directory is missing.
- Restore the missing `patient` directory from the incremental backup.
- List and verify the `patient` directory to verify the restoration was successful.
- Create two new files in the `patient` directory.
- Create an incremental backup on Monday and verify that the new `patient` files have been properly backed up.

Finally, your team will conclude this exercise by reviewing file attributes as a result of the backup process.

Lab Environment

If you are not already logged on, log into this module's lab environment using the following credentials:

- Username: `sysadmin`
- Password: `cybersecurity`

Instructions

1. Move into the `~/Documents/epscript/testenvir` directory and list the contents.

2. In your `~/Documents/epscript` directory, create the level 0 backup of the `testenvir` directory, which contains the `doctor`, `patient`, and `treatment` directories.
 - Change back into the `epscript` directory:
 - Run `cd ../`
 - Create the level 0 backup of the `testenvir` directory.
3. View and verify the contents of the **level 0** backup by using `tar`.
 - What is the status of the files in the backup?
 - What does `D` indicate?
 - What does `x` indicate?
4. Next, we'll simulate a natural disaster or cyberattack by removing the `patient` directory.
 - From the `~/Documents/epscript` directory:
 - Run `rm -r testenvir/patient/`.
 - Verify that the `patient` directory is removed.
 - Run `ls -l testenvir/`.
- Your output should look similar to the following. Notice that the `patient` directory is missing:

Unset

```
total 8
drwxr-xr-x 2 sysadmin sysadmin 4096 Jul 14 10:14 doctor
drwxr-xr-x 3 sysadmin sysadmin 4096 Jul 14 10:14 treatment
```

5. Restore the missing `patient` directory from the `epscript_back_sun.tar` backup to the `~/Documents/epscript/testenvir/patient/` directory.
 - Make sure you're in the `epscript` directory:
 - Run `cd ~/Documents/epscript`.
 - Restore the missing **patient** directory.
 - Verify that the files have been added to the `testenvir/patient` directory successfully.
 - Run `ls -l testenvir/`.
 - If successful, the missing `patient` directory should be properly restored from the archive.
6. Before we create an incremental backup, we'll add new files. Let's add a couple of files to the `patient` directory:
 - First, ensure you're in the `cd ~/Documents/epscript/testenvir/patient/` directory.
 - Run `cd ~/Documents/epscript/testenvir/patient/`.
 - Create a few arbitrary files:
 - For example: `touch patient.0a.txt patient.0b.txt`.
 - Verify the files have been created
 - Run `ls -l`.
 - Output should look similar to the following. Notice the two new files `patient.0a.txt` and `patient.0b.txt` have been successfully created:

Unset

```
total 284
-rw-r--r-- 1 sysadmin sysadmin      0 Aug 13 12:27 patient.0a.txt
-rw-r--r-- 1 sysadmin sysadmin      0 Aug 13 12:27 patient.0b.txt
-rw-r--r-- 1 sysadmin sysadmin 6236 Jul 14 10:14 patients.10.csv
-rw-r--r-- 1 sysadmin sysadmin 6250 Jul 14 10:14 patients.11.csv
```

7. Assume it's Monday. Now, we'll create an incremental backup that will include our newly created patient documents.
 - Change back to the `~/Documents/epscript` directory:
 - Run `cd ~/Documents/epscript`.
 - Create an incremental backup for Monday.
 - List the contents of the `epscript_back_mon.tar` incremental backup and verify that the new files have been archived.
 - What is the status of the files in the incremental backup?

For additional reference, consider the following:

- [Manpages: backup-manager](#)
- [Amanda.org](#)

Bonus Review Questions

1. Briefly describe the difference between a full and incremental backup.
2. If you have a backup schedule of Monday, Tuesday, Wednesday, Thursday, and Friday:
 - On what day would you schedule a level 0 backup to be done?
 - In what order should the backups be applied to restore a system that was completely lost after an attack?
3. What command would you use to create a level 0 backup of `archive/home/user1`?
4. What command would you use to list the contents of an incremental backup?
5. After listing the contents of an incremental backup, what do the following letters indicate?
 - **Y** indicates that _____.
 - **N** indicates that _____.
 - **D** indicates that _____.